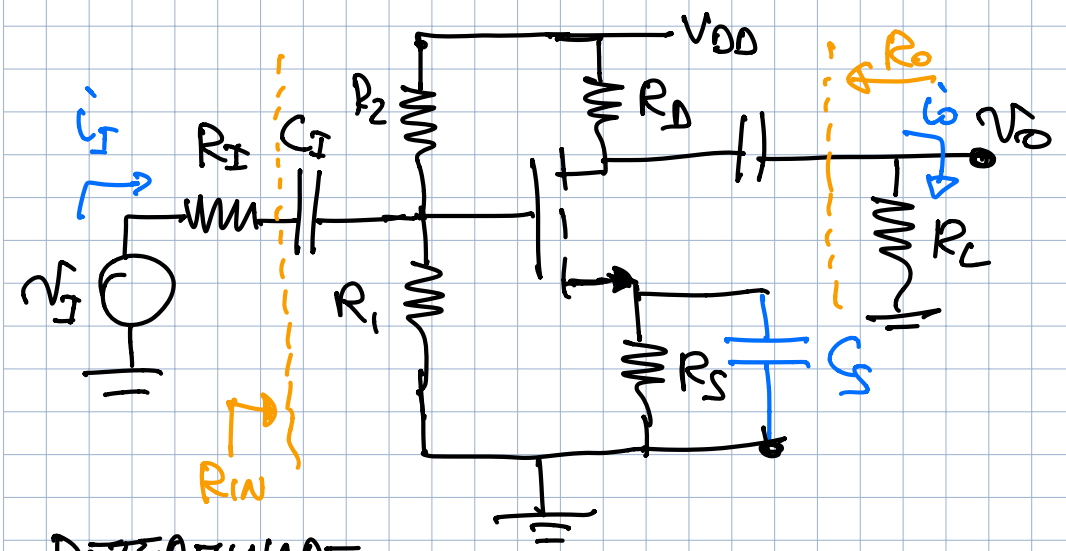
ESERCIZIO A2

AMPLIFICATORE A SOURCE COMUNE



$V_{DD} = 15V$

$V_{TN} = 1V$,

$K_n = 2{,}0\ mA/V^2$

$R_1 = 1\,M\Omega$

$R_2 = 1\,M\Omega$

$R_I = 1\,K\Omega$

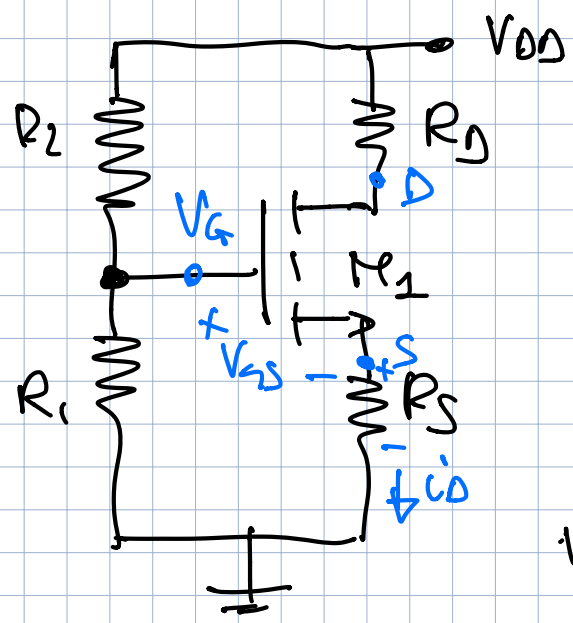
$R_L = 5\,K\Omega$

$\lambda = 0{,}01\,V^{-1}$

DETERMINARE

1) R_D, R_S PER AVERE $I_D = 4\,mA$, $V_{DS} = 6\,V$
2) RESISTENZE DI INGRESSO e USCITA
3) GUADAGNO DI TENSIONE $A_V = \frac{v_O}{v_I}$
4) GUADAGNO DI CORRENTE $A_I = \frac{i_O}{i_I}$
5) RIPETERE 1–4 SENZA C_S.

ANALISI DC (CALCOLO PUNTO DI LAVORO)



$$V_G = V_{DD} \cdot \frac{R_1}{R_1 + R_2} = 7{,}5\ V$$

$V_G > V_{TN} \Rightarrow$ MOSFET "ON"

HP MOSFET SATURAZIONE

$$I_D = \frac{K_n}{2}(V_{GS} - V_{TN})^2$$

$$V_{GS} = V_{TN} \pm \sqrt{\frac{2 I_D}{K_n}} = \begin{cases} 3\,V & \text{OK} \\ -1\,V & \text{NO } < V_{TN} \end{cases}$$

$\Rightarrow \quad V_S = V_G - V_{GS} = 4{,}5\,V$

$\Rightarrow \quad R_S I_D = V_S \quad \rightarrow \quad R_S = \dfrac{V_S}{I_D} = \dfrac{4{,}5\,V}{4\,mA} = 1{,}125\,K\Omega$

$V_D = V_S + V_{DS} = 10{,}5\,V$

$\Rightarrow \quad V_{DD} - R_D I_D = V_D \quad \Rightarrow \quad R_D = \dfrac{V_{DD} - V_D}{I_D} =$

$= \dfrac{15\,V - 10{,}5\,V}{4\,mA} = 1{,}125\,K\Omega$

$V_{DS} \geq V_{GS} - V_{TN}$?

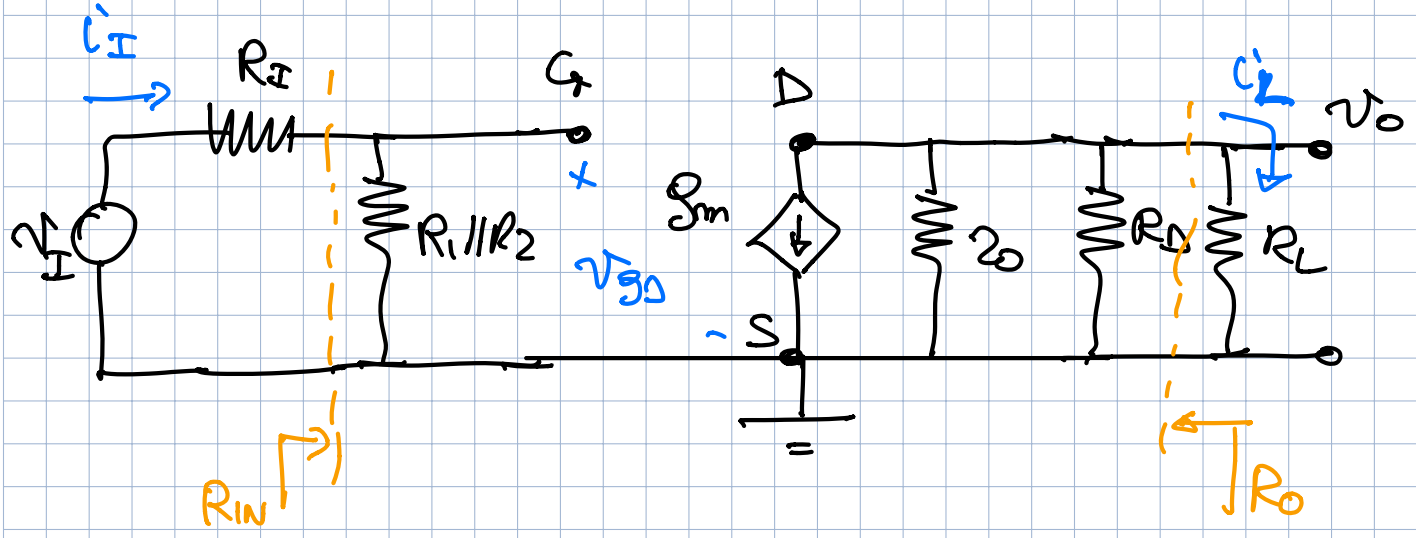
$6\,V > 3\,V - 1\,V = 2\,V$ OK MOSFET IN SATURAZIONE

PARAMETRI PICCOLO SEGNALE:

$$g_m = \frac{2 I_D}{V_{GS} - V_{TN}} = \frac{8\ mA}{2\ V} = 4{,}0\ mS$$

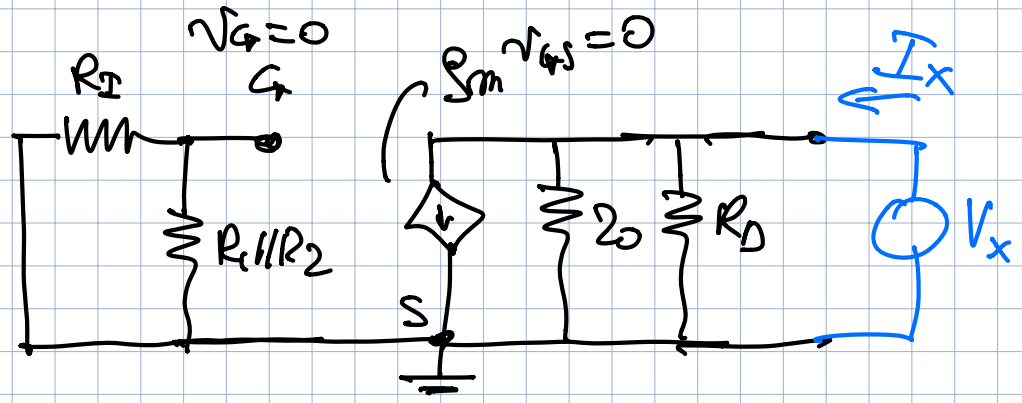
$$r_o = \frac{\frac{1}{\lambda} + V_{DS}}{I_{DQ}} = \frac{100 + 6{,}0\ V}{4\ mA} = 26{,}5\ K\Omega$$

ANALISI AL PICCOLO SEGNALE



$R_{IN} = R_1 // R_2 = 500\,K\Omega$

R_o ?



$v_G = 0$

$\Rightarrow g_m v_{GS} = 0 \quad \Rightarrow V_x = I_x (r_o // R_D)$

$$\Rightarrow R_o = \frac{V_x}{I_x} = r_o // R_D = 1{,}08\ K\Omega$$

$$A_v = \frac{v_o}{v_I} = \frac{v_o}{v_G} \cdot \frac{v_G}{v_I} = -3{,}54$$

$$\frac{v_G}{v_I} = \frac{R_1 // R_2}{R_I + R_1 // R_2} = 0{,}988$$

N.B

$v_G = v_{GD}$

$$\frac{v_o}{v_G} = -g_m \underbrace{r_o // R_D // R_L}_{888{,}2\ \Omega} = -3{,}55$$

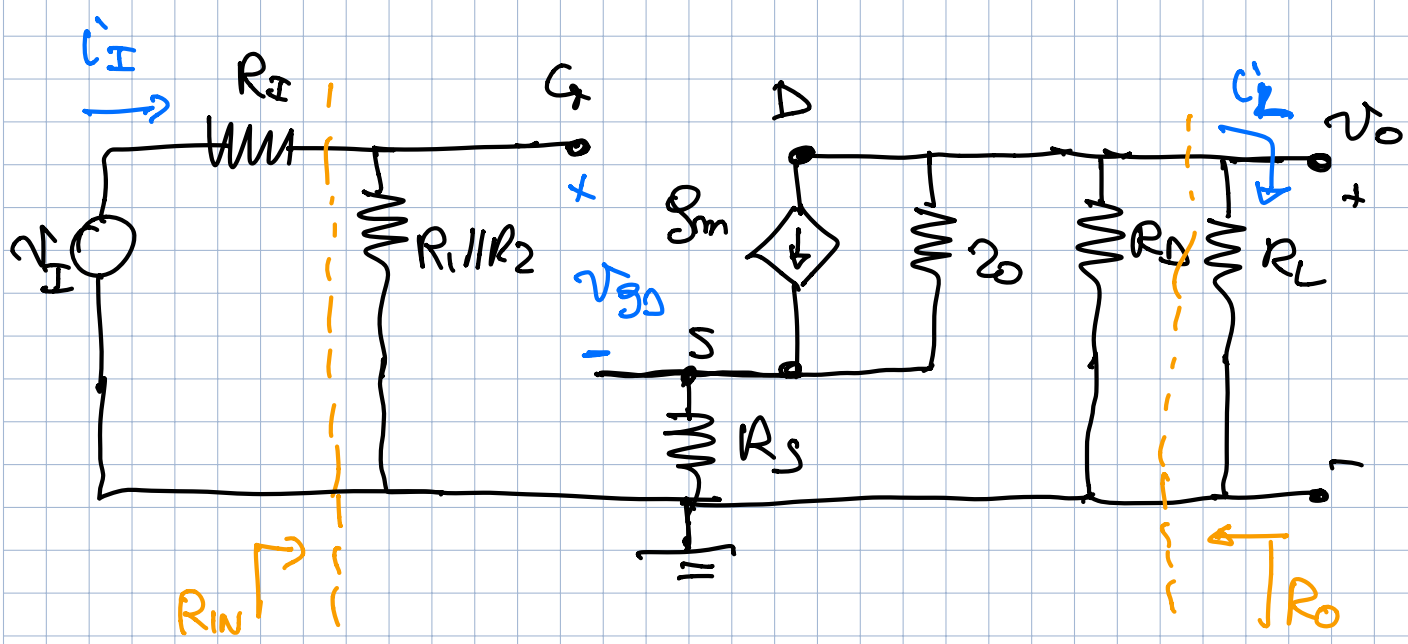
$$A_i = \frac{i_o}{i_I} = \frac{v_o}{R_L} \cdot \frac{1}{\frac{v_I}{R_I + R_1 // R_2}} = \frac{v_o}{v_i} \cdot \underbrace{\frac{R_I + R_1 // R_2}{R_L}}_{100{,}2}$$

$$A_i = A_v \cdot \frac{R_I + R_1 // R_2}{R_L} = -355$$

COSA SUCCEDE SE TOLGO C_S ?

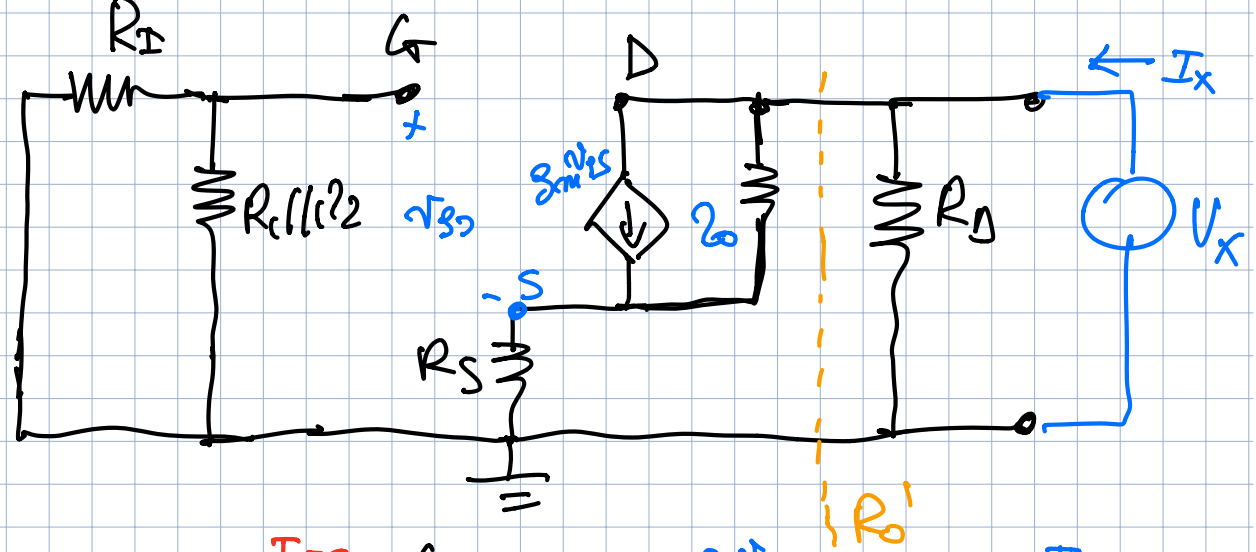
PER LA POLARIZZAZIONE NON CAMBIA NULLA!
MA NELLA AC CAMBIA.

ANALISI AL PICCOLO SEGNALE SENZA C_S



$R_{IN} = R_1 // R_2 = 500\,K\Omega$ (come prima)

R_O ?



$R_O = R_D // R_O'$

$I_G = 0$

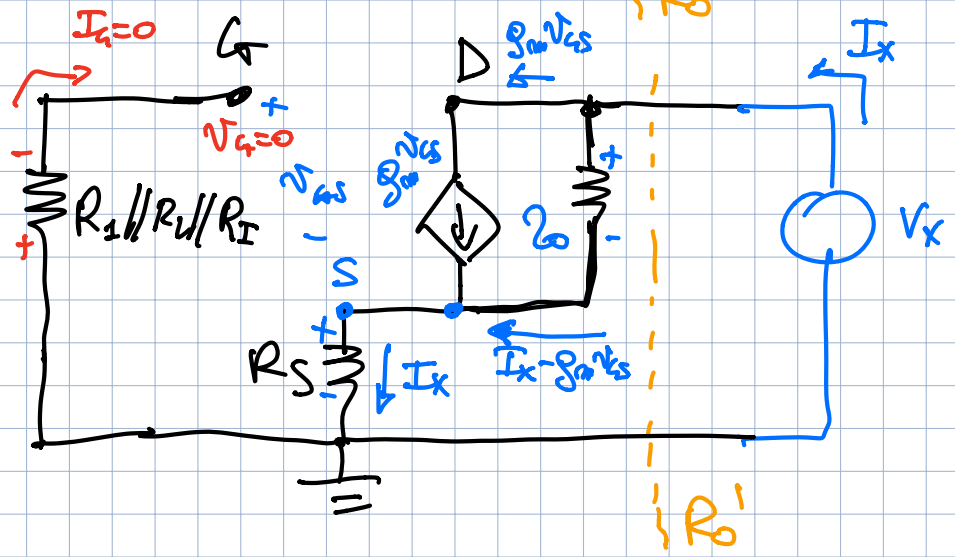
$v_G = 0$

$v_S = I_x R_S$

$v_{GS} = v_G - v_S$

$= -v_S$

$= -I_x R_S$



$$V_x = (I_x - g_m v_{GS})\, r_o + I_x R_S$$

$$V_x = (I_x + g_m I_x R_S)\, r_o + I_x R_S$$

$$R_o' = \frac{V_x}{I_x} = \underbrace{(1 + g_m R_S)\, r_o}_{\text{IN GENERE È} \gg R_S} + R_S$$

$$R_o' = (1 + 4\,mS \cdot 1{,}125\,K\Omega) \cdot 26{,}5\,K\Omega + 1{,}125\,K\Omega$$

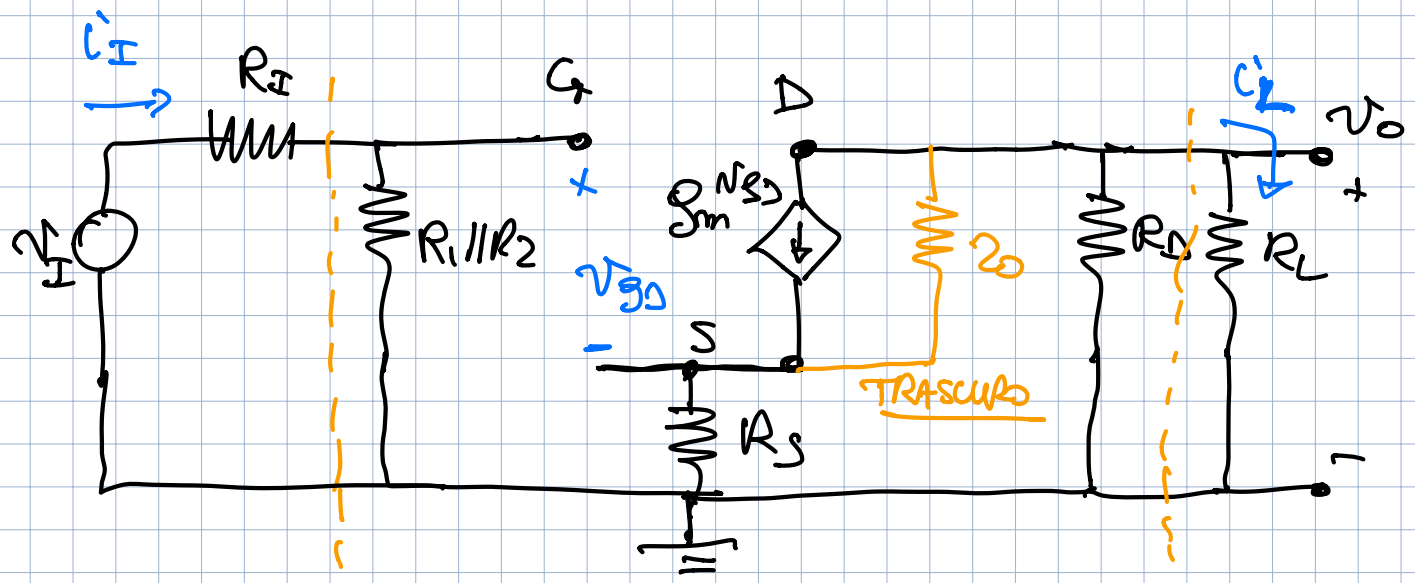
$$\simeq 145{,}75 + 1{,}125\,K\Omega = 146{,}875\,K\Omega$$

$$\Rightarrow R_o = R_D // R_o' = 1{,}116\,K\Omega$$

SPESSO SI CONSIDERA $R_o' = \infty$

$\Rightarrow R_o = R_D$!

VA GIUSTIFICATA



$$A_v = \frac{v_o}{v_I} = \frac{v_o}{v_g} \cdot \frac{v_g}{v_I}$$

$$v_o = -g_m v_{GS}\, R_D // R_L$$

$$v_g = v_{GS} + R_S g_m v_{GS} = (1 + g_m R_S)\, v_{GS}$$

$$v_{GS} = \frac{v_g}{1 + g_m R_S} \quad \Rightarrow v_o = -\frac{g_m R_D // R_L}{1 + g_m R_S}\, v_g$$

$$\Rightarrow \frac{v_o}{v_g} = -\frac{g_m R_D \| R_L}{1 + g_m R_S} = \frac{-3{,}67}{5{,}5} = \underline{\underline{-0{,}667}}$$

$$\frac{v_g}{v_I} = \frac{R_1 \| R_2}{R_I + R_1 \| R_2} = 0{,}998$$

$$\Rightarrow \underline{\underline{A_v = -0{,}67}}$$

$$\underline{A_I = \frac{i_o}{i_I} = A_v \cdot \frac{R_I + R_1 \| R_2}{R_L} = -66{,}7}$$